

FlowOps — Architecture & Design

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FlowOps — Architecture & Design Document

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Appendix A — Review Provenance

title: “FlowOps — Architecture & Design Document” subtitle: “Governed self-service provisioning: the governed actuation layer” author: “Brian Duffy” date: “June 8, 2026”

FlowOps — Architecture & Design Document

Status: Plan approved (Eng + CEO + Design reviews cleared). Pre-implementation. **Version:** 1.0 (v1 scope) **Document owner:** Brian Duffy

1. Executive Summary

FlowOps is an open-source **governed self-service provisioning platform** — *the governed actuation layer*. It lets a person request infrastructure, pass a policy and budget gate, and have it actually **built** — with controls (approval, budget, policy, audit) that never come off, and with agents and automation able to do the work *inside* the guardrails.

It begins as a refined work-intake Kanban proof of concept and grows into something sharper: **AI-native ITSM**. ITSM's skeleton — `request → task → action`, cleanly linked — is genuinely good. Its flesh — the human bureaucracy that fulfills every task — is pure toil. ServiceNow won in 2005 because Remedy made you waste your life on care-and-feeding. In 2026 the toil worth killing is the *human-in-every-step* fulfillment. FlowOps keeps the elegant `request → task → action` model and makes a task's fulfiller a human, an automation, or an AI agent — while governance stays constant no matter who or what does the work.

The v1 wedge is **sandbox vending**: a developer (or AI) sandbox that is requested, gated, provisioned, and auto-torn-down on a TTL — proving the whole governed-actuation loop end to end. It lives in the unserved middle between ClickOps (manual, ungoverned, doesn't scale) and full CI/CD pipelines (overkill, get in the way).

This document captures the objectives, the chosen approach, the engineering architecture, the design system, and the strategy, as decided across four structured reviews (product, engineering, strategy, design).

2. Objectives

1. **Kill one unit of toil for real.** A non-author colleague requests a dev sandbox, it passes an approval + cloud-enforced budget gate, real infrastructure is provisioned and auto-torn-down on TTL — with a full audit trail — without anyone hand-running infrastructure-as-code.
2. **Make governance first-class, not bolted on.** Every fulfilled task is gated and audited. The controls are the product.
3. **Be genuinely open source and embeddable.** A single organization can run the complete governed-actuation loop for free; the platform is designed to be embedded in other internal platforms.
4. **Earn the right to the bigger vision.** Prove one concrete loop, then generalize to the agentic workflow engine and the embeddable actuation core.

2.1 POSITIONING

The category FlowOps claims is "**governed self-service provisioning — the governed actuation layer.**" The headline is governance + actuation as one loop ("a request passes a gate and gets *built*, and the controls never come off"). The sandbox is the concrete demo *under* the category; "AI-native ITSM" is the vision/roadmap narrative.

This matters because category sets the comparison set. Called a "sandbox tool," FlowOps is a feature next to Backstage's scaffolder. Called "the governed actuation layer," it defines a lane it can own.

2.2 LANDSCAPE

Space	Examples	Why FlowOps is different
OSS ITSM	GLPI, Zammad, iTop, OTOBO, osTicket	Those track tickets. FlowOps <i>actuates</i> and governs.
IDP / platform	Backstage, Port, Cortex, Humanitec	Those are catalogs + scaffolding, often BYO-everything. FlowOps is a batteries-included governed loop .
Actuation	Crossplane, AWS Service Catalog, Spacelift, env0	Those provision. FlowOps provisions <i>inside</i> gates with audit, as one loop.
Commercial ITSM	ServiceNow, Atlassian JSM	FlowOps is the AI-native, open-source reframe.

The seam almost nobody owns is the *middle*: intake + governance + actuation as one loop, where the gate, the budget check, the policy enforcement, and the provisioning are the same system, and agents do the labor inside the guardrails.

3. Scope

3.1 IN SCOPE (V1)

The minimal loop that proves the thesis:

`request → task → action`, with one blueprint (**dev sandbox**), one gate set (approval

- cloud-enforced budget), one real actuator (OpenTofu in a container job), TTL teardown, and an append-only audit trail.

3.2 EXPLICITLY NOT IN SCOPE (V1)

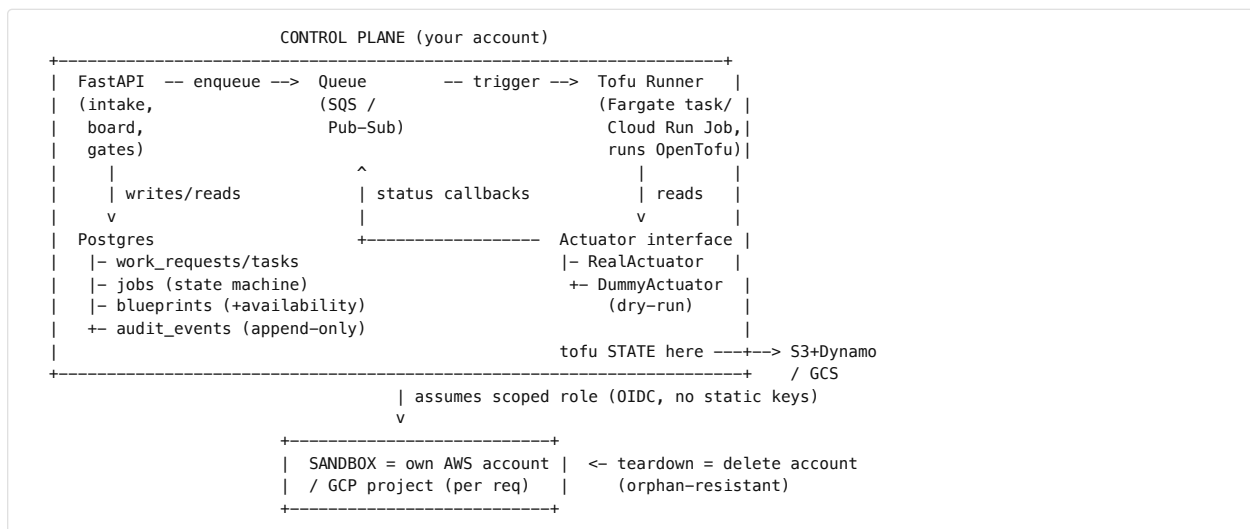
Deferred	Rationale
AI sandbox blueprint	Fast-follow (#2 blueprint); proves the actuator seam. Needs definition first.
Agent fulfiller	The emotional center of the project, deliberately deferred to protect credibility — v1 proves the <i>loop and gates</i> , not yet the thesis.
Plugin registry / Gate & Fulfiller interfaces	Extract on the second real instance, not before.
Live cost estimation (Infracost)	Cloud-enforced budget replaces it for v1.
Warm pool of pre-vended accounts	Perf optimization for when on-demand vending latency hurts.
Multi-tenancy, SSO, hosted runners	Commercial / open-core layer, post-v1.

3.3 HONEST TENSION

The most exciting part of the vision — agents doing the work inside the gates — is *not* in v1. v1 proves the governance + actuation loop with human-approval + automation only. This is the right call: it keeps the “it actually gets built” promise honest. The agent fulfiller is fast-follow #1.

4. Architecture

4.1 SYSTEM OVERVIEW



4.2 EXECUTION MODEL — DURABLE QUEUE + SERVERLESS CONTAINER JOB

`tofu apply` / `destroy` is a multi-minute operation that can fail halfway. A synchronous HTTP handler cannot own it. FlowOps uses a **durable queue (SQS / Pub-Sub) feeding a serverless container job** (AWS Fargate task / GCP Cloud Run Job) that runs OpenTofu.

- No 15-minute function ceiling; crash-safe via queue redelivery.

- A `jobs` table is the state machine. **One job path serves both apply and destroy.**
- Required mechanics: idempotency keys, per-sandbox locks, job leases, and “apply-succeeded-but-callback-failed” reconciliation.
- `status()` reads the jobs table — it is not a live cloud poll.

```
JOB STATE MACHINE (one path for apply AND destroy):
  queued -> planning -> awaiting_gate -> applying -> succeeded
      |           |           |           |           |
      v           | (deny)    |           |           |
      v           v           v           v (ttl)
    failed    rejected    failed<---+    destroying -> destroyed
                                +--- reconcile / orphan-sweep
```

4.3 ISOLATION — ACCOUNT/PROJECT PER SANDBOX

Each sandbox is its **own AWS account** (Organizations) or **GCP project**. The runner assumes a scoped role via OIDC — no stored static keys.

- **Teardown = delete the account/project.** This makes the two scariest problems structurally safe: a failed apply or a missed orphan is recovered by deleting the whole account/project, not by hoping a tag-sweep caught everything.
- Orphan-*resistant*, not instant or perfect: AWS account closure has a ~90-day window, GCP project deletion a ~30-day recovery window, plus billing tails — design for the tail.
- OpenTofu **state lives in the control-plane account**, encrypted, with secrets redacted and access isolated.

The first cloud (AWS vs GCP) is decided in a **week-0 spike** that builds the apply/destroy/orphan test against both and picks the factory that is demonstrably safe and fast (GCP project vending is roughly one API call; AWS account factory is heavier OU/SCP/Control Tower ceremony).

4.4 THE ACTUATOR SEAM (THE ONLY ABSTRACTION IN V1)

```
class Actuator(Protocol):
    def plan(self, intent: Intent) -> Preview: ...      # pre-flight: what + est. cost
    def apply(self, intent: Intent) -> Handle: ...      # returns state-locating handle
    def destroy(self, handle: Handle) -> Result: ...
    def status(self, handle: Handle) -> Status: ...
```

Two v1 implementations: `RealActuator` (the container job) and `DummyActuator` (dry-run). The dummy is also the test double *and* the placeholder for not-yet-live capabilities — it lets the whole loop be built, demoed, and tested before the cloud is wired. Capability flags (`available` | `preview` | `dry-run-only`) let the UI build around absent features, but absent capabilities are shown as absent/preview, never silently faked.

Gates (approval, budget) and `fulfiller_type` stay concrete — no plugin registry until a real second instance forces the interface shape.

4.5 BUDGET GATE — CLOUD-ENFORCED, NOT DECLARED

The budget gate is a **hard ceiling the cloud enforces**, not a user promise. At provision time, FlowOps sets a real AWS Budget + SCP / GCP budget + quota + service allowlist on the sandbox account. A declared “this fits under \$X” is governance theater; the cloud-enforced ceiling is the only real control — and it removes the need for live cost estimation in v1.

4.6 DOMAIN MODEL

`request -> task -> action`, kept thin (no ITSM ceremony — no SLAs, categories, queues). The task *is* the provisioning job; the action *is* the actuator call. This keeps the thesis structurally true so the agent-fulfiller future slots in without a rename.

Entity	Key fields
<code>Blueprint</code>	<code>`id, type(dev</code>
<code>Task</code>	<code>`fulfiller_type(human</code>
<code>Gate</code>	<code>`type(approval</code>
<code>ActuatorHandle</code>	<code>state_backend_ref, workspace, outputs{ }, status</code>
<code>AuditEvent</code>	<code>type, actor, task_id, correlation_id, payload, ts (append-only)</code>

5. Engineering Considerations

5.1 FAILURE MODES

Failure	Covered by	Silent?
Mid-apply failure -> orphaned spend	account/project deletion + reconcile sweep + test	No — job->failed, owner alerted
Worker crash mid-apply -> double-apply	idempotency key + per-sandbox lock + redelivery test	No
Budget bypass -> runaway spend	cloud-enforced budget + SCP + quota (hard ceiling)	No — cloud kills it
Wrongful teardown -> data loss	grace period + owner notice + never-destroy guard + fail-closed	No
Missing cost/budget config	fail-closed (block apply)	No
Control-plane compromise	state encryption + secret redaction + access isolation	Residual risk — flagged, monitored

Global invariant: fail closed. Every gate and the teardown guard default to DENY on ambiguity — a missing cost estimate, an unknown role, an unparseable handle blocks, never proceeds.

5.2 TESTING STRATEGY

Full failure-path coverage. The failure tests *shape* the job model — the state machine and its failure tests are written first, before happy-path actuation. `DummyActuator` drives fast, deterministic unit and integration tests with zero cloud credentials; integration tests against real OpenTofu (localstack / a throwaway project) cover state locking and cloud-failure semantics the dummy cannot prove. Required tests include: mid-apply failure + reconcile, crash redelivery (no double-apply), budget at/over/under cap, missing-cost-preview = fail closed, teardown guardrails, and quota exhaustion.

5.3 SECURITY & SUPPLY CHAIN

- OIDC scoped-role assumption; bootstrap trust, per-account role creation, least-privilege policy, and revocation/rotation are part of the week-0 spike.
- Curated OpenTofu modules are pinned by ref, with a provider lockfile + checksums and a review policy. A “curated” module is still supply chain.
- Append-only audit with correlation IDs, actor-identity provenance, immutable job inputs, and redaction rules.

5.4 TOOLING CHOICES (BORING BY DEFAULT)

OpenTofu (not Terraform — MPL licensing fits an OSS project), Python/FastAPI control plane, Postgres, a managed queue, object-store state backend. Innovation tokens are reserved for the agent-fulfiller layer, not the plumbing.

5.5 DISTRIBUTION

AGPL-3.0 core in an OSS repo. `docker compose up` runs the **app** in minutes; the **actuation** half requires guided cloud wiring (it cannot be zero-config). CI via GitHub Actions builds/publishes container images on tag, with GitHub Releases for versioned artifacts. A hosted/managed offering is deferred until after v1 delights a self-host user.

6. Design

6.1 DESIGN SYSTEM — NYS DESIGN SYSTEM (MANDATORY)

FlowOps targets a **New York State government** context. The UI conforms to the **NYS Design System** (USWDS-derived) and is built on `---nys-color-*` design tokens (CSS custom properties) so it is themeable:

Token	Value	Use
<code>--nys-color-theme</code>	<code>#154973</code>	Primary state blue; active state, section accents
<code>--nys-color-link</code> / <code>--nys-color-focus</code>	<code>#004dd1</code>	Interactive, links, focus rings
<code>--nys-color-success</code>	<code>#1e752e</code>	Passed / healthy
<code>--nys-color-text</code>	<code>#1b1b1b</code>	Body text
<code>--nys-color-surface</code>	<code>#ffffff</code>	Page surface
Environment theme	<code>#233f2b</code>	Alternate theme

The proof-of-concept's orange admin accent is **superseded by the NYS theme blue**. Typography is an accessible government sans (Public Sans family), never `system-ui`.

Accessibility is mandatory (WCAG 2.1 AA / Section 508): visible focus rings, $\geq 4.5:1$ body contrast, 44px touch targets, full keyboard navigation, ARIA landmarks, and screen-reader labels on the status pill and lifecycle stepper. Status is never conveyed by color alone — color is always paired with icon + text.

6.2 INFORMATION ARCHITECTURE — SANDBOX LIFECYCLE VIEW

Reading order, by priority: (1) identity + current status; (2) the lifecycle stepper (where is this, right now); (3) governance (budget ceiling + Guardrails + Approve/Deny); (4) TTL countdown + Extend; (5) provisioned outputs; (6) append-only audit timeline. Status legibility is the number-one job — never plain “Status: applying” text.

6.3 APPROVED MOCKUP

The approved build reference for the sandbox lifecycle detail view (NYS Design System palette, calm layout, per-event audit iconography, guardrails panel that makes “controls never come off” literally visible):

 Approved sandbox lifecycle detail view (NYS Design System palette)

An alternate top-navigation treatment of the same screen:

 Alternate layout — persistent top nav + global search

For contrast, an earlier exploration in the original proof-of-concept palette (orange accent), before the NYS Design System constraint reshaped the direction:

 Early exploration — original PoC orange palette, bolder state machine

6.4 INTERACTION-STATE COVERAGE

Surface	Loading	Empty	Error	Success	Partial
Board card	skeleton + “Queued...”	n/a	red pill “Apply failed” + View error	green “Running” + TTL chip	amber “Applying... {step}”
Lifecycle detail	stepper skeleton	n/a	failed step red, audit shows error, Retry/Destroy	full stepper, outputs, TTL ticking	mid-stepper pulsing, outputs hidden until Running
Gate approval	“Loading cost preview...” (Approve disabled)	n/a	“Cost preview unavailable — cannot approve” (fail-closed)	Approve enabled, cost + budget shown	read-only awaiting banner for non-approvers
Failure / expiry	—	—	mid-apply failure: red banner + orphan-reconcile + owner alert	clean teardown: “Destroyed” terminal state + final audit	TTL near-expiry: amber countdown + prominent Extend

7. Strategy

Decision	Choice
License	AGPL-3.0 + commercial dual-license (requires a CLA from day one)
Open-core line	Generous core — full single-org governed-actuation loop is OSS; sell multi-tenancy, SSO/SAML/SCIM, advanced policy, hosted runners, audit retention/compliance, support
Positioning	"Governed self-service provisioning — the governed actuation layer." Sandbox = demo; AI-native ITSM = vision narrative

The AGPL + commercial structure monetizes proprietary embedding directly, which is why the feature core can be generous: the OSS version is a complete production tool for one organization, not a toy.

8. Build Plan

8.1 PARALLEL LANES

Lane	Modules	Depends on
A — control plane	<code>api/</code> , <code>models/</code> (PoC port, domain model, jobs table)	—
B — actuation	<code>actuators/</code> , jobs worker, queue wiring	A's jobs table (Dummy path early; Real path needs C)
C — cloud factory (week-0 spike)	<code>infra/</code> (account/project vending, budget/SCP, OIDC trust)	—

Launch A + C in parallel (C is the spike). B starts against `DummyActuator` immediately, integrating the Real path when C lands. A and B share the `jobs` table — one owner defines that schema first.

8.2 SEQUENCED TASKS (HIGHLIGHTS)

- Week-0 spike (blocking):** prove apply/destroy/orphan-detect + OIDC trust + budget/SCP against both GCP project and AWS account; pick the first cloud.
- Bring the PoC into the repo; module layout `api/ actuators/ gates/ jobs/ audit/ infra/`.
- Jobs/state-machine table; queue + container-job runner.
- Actuator interface + Dummy + Real; approval + cloud-enforced budget gate.
- TTL teardown via the same job path; append-only audit.
- Full failure-path test suite (written with the state machine).
- Licensing: `LICENSE` (AGPL-3.0), `COMMERCIAL.md`, CLA bot, documented open-core line.
- Design: `DESIGN.md` (NYS token mapping), build the lifecycle view from the approved mockup, implement all interaction states, WCAG 2.1 AA pass.

9. Open Risks

- Control-plane blast radius:** state + secrets concentrate in the control plane. Mitigated (encryption, redaction, isolation), not eliminated — monitor.
- Account-factory weight:** the cloud factory could become the product before sandbox vending exists. The week-0 spike timeboxes and de-risks this.
- "Out of the box" honesty:** the app runs out of the box; actuation requires guided cloud setup. Positioned as a batteries-included governed loop vs a BYO toolkit — a strength, framed honestly, not a retreat.

Appendix A — Review Provenance

This plan was produced and pressure-tested across four structured reviews:

- Product (office hours):** reframed work-intake PoC into AI-native ITSM / governed actuation; locked the sandbox wedge and "A + seam toward B" approach.

- **Engineering review:** locked execution model, isolation, the Actuator seam + dummy, and full failure-path testing. An independent cross-model pass (Codex, GPT-5.5) surfaced 23 challenges — integrated or routed.
- **CEO / strategy review:** locked license (AGPL + commercial), the generous open-core line, and positioning.
- **Design review:** rated the new provisioning UI 3/10 -> 8/10; surfaced the NYS Design System + accessibility constraint; approved the lifecycle mockup; specified all interaction states.

All decisions were made by the project owner; cross-model input was advisory.